Automotive AI Assistant using RAG

Objective: To build an AI-powered assistant that answers vehicle-related queries such as specifications, service schedules, and recommendations.

Technologies Used: Python, Sentence Transformers, FAISS, NumPy

System Architecture:
User Query → Embedding → FAISS Search → Retrieve Data → Generate Answer

Dataset: Synthetic dataset including vehicle models, service details, and warning messages.

Embeddings: Embeddings convert text into numerical vectors, enabling similarity comparison between queries and stored data.

Cosine Similarity: Measures similarity between vectors. Higher value means more similar meaning.

RAG (Retrieval-Augmented Generation): Combines retrieval of relevant data with answer generation.

Importance of Grounding: Ensures accurate answers in automotive systems where incorrect information can be risky.

Hallucination Causes: Lack of data, weak prompts, and over-generalization.

Mitigation: Use retrieved context, restrict responses, and fallback when data is missing.

Features: Semantic search, RAG-based answering, vehicle recommendation system.

Conclusion: The system demonstrates how AI can assist in automotive queries with accurate and relevant responses.